Incorporating Copying Mechanism in Sequence-to-Sequence Learning

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Abstract

We address an important problem in sequence-to-sequence (Seq2Seq) learning referred to as copying, in which certain segments in the input sequence are selectively replicated in the output sequence. A similar phenomenon is observable in human language communication. For example, humans tend to repeat entity names or even long phrases in conversation. The challenge with regard to copying in Seq2Seq is that new machinery is needed to decide when to perform the operation. In this paper, we incorporate copying into neural network-based Seq2Seq learning and propose a new model called COPYNET with encoder-decoder structure. COPYNET can nicely integrate the regular way of word generation in the decoder with the new copying mechanism which can choose subsequences in the input sequence and put them at proper places in the output sequence. Our empirical study on both synthetic data sets and real world data sets demonstrates the efficacy of COPYNET. For example, COPYNET can outperform regular RNN-based model with remarkable margins on text summarization tasks.

1 Introduction

Recently, neural network-based sequence-to-sequence learning (Seq2Seq) has achieved remarkable success in various natural language processing (NLP) tasks, including but not limited to Machine Translation (Cho et al., 2014; Bahdanau et al., 2014), Syntactic Parsing (Vinyals et al., 2015b), Text Summarization (Rush et al., 2015) and Dialogue Systems (Vinyals and Le, 2015). Seq2Seq is essentially an encoder-decoder model, in which the encoder first transform the input sequence to a certain representation which can then transform the representation into the output sequence. Adding the attention mechanism (Bahdanau et al., 2014) to Seq2Seq, first proposed for automatic alignment in machine translation, has led to significant improvement on the performance of various tasks (Shang et al., 2015; Rush et al., 2015). Different from the canonical encoder-decoder architecture, the attention-based Seq2Seq model revisits the input sequence in its raw form (array of word representations) and dynamically fetches the relevant piece of information based mostly on the feedback from the generation of the output sequence.

In this paper, we explore another mechanism important to the human language communication, called the "copying mechanism". Basically, it refers to the mechanism that locates a certain segment of the input sentence and puts the segment into the output sequence. For example, in the following two dialogue turns we observe different patterns in which some subsequences (colored blue) in the response (R) are copied from the input utterance (I):

I: `Hello Jack, my name is Chandralekha.`
R: `Nice to meet you, Chandralekha.`

I: `This new guy doesn't perform exactly as we expected.`
R: `What do you mean by "doesn't perform exactly as we expected"?`

Both the canonical encoder-decoder and its variants with attention mechanism rely heavily on the representation of "meaning", which might not be sufficiently inaccurate in cases in which the system needs to refer to sub-sequences of input like entity names or dates. In contrast, the copying mechanism is closer to the rote memorization in language processing of human being, deserving a different modeling strategy in neural network-based models. We argue that it will benefit many Seq2Seq tasks to have an elegant unified model that can accommodate both understanding and rote memorization. Towards this goal, we propose COPYNET, which is not only capable of the regular generation of words but also the operation of copying appropriate segments of the input sequence. Despite the seemingly "hard" operation of copying, COPYNET can be trained in an end-to-end fashion. Our empirical study on both synthetic datasets and real world datasets demonstrates the efficacy of COPYNET.

2 Background: Neural Models for Sequence-to-sequence Learning

Seq2Seq Learning can be expressed in a probabilistic view as maximizing the likelihood (or some other evaluation metrics (Shen et al., 2015)) of observing the output (target) sequence given an input (source) sequence.

2.1 RNN Encoder-Decoder

RNN-based Encoder-Decoder is successfully applied to real world Seq2Seq tasks, first by Cho et al. (2014) and Sutskever et al. (2014), and then by (Vinyals and Le, 2015; Vinyals et al., 2015a). In the Encoder-Decoder framework, the source sequence $X = [x_1, ..., x_{T_S}]$ is converted into a fixed length vector $\mathbf{c}$ by the encoder RNN, i.e.

$$\mathbf{h}_t = f(x_t, \mathbf{h}_{t-1}); \quad \mathbf{c} = \phi(\{\mathbf{h}_1, ..., \mathbf{h}_{T_S}\}) \quad (1)$$

where $\{\mathbf{h}_t\}$ are the RNN states, $\mathbf{c}$ is the so-called context vector, f is the dynamics function, and ϕ summarizes the hidden states, e.g. choosing the last state $\mathbf{h}_{T_S}$. In practice it is found that gated RNN alternatives such as LSTM (Hochreiter and Schmidhuber, 1997) or GRU (Cho et al., 2014) often perform much better than vanilla ones.

The decoder RNN is to unfold the context vector $\mathbf{c}$ into the target sequence, through the following dynamics and prediction model:

$$\begin{aligned} \mathbf{s}_t &= f(y_{t-1}, \mathbf{s}_{t-1}, \mathbf{c}) \\ p(y_t|y_{<t}, X) &= g(y_{t-1}, \mathbf{s}_t, \mathbf{c}) \end{aligned} \quad (2)$$

where $\mathbf{s}_t$ is the RNN state at time t, y_t is the predicted target symbol at t (through function $g(\cdot)$) with $y_{<t}$ denoting the history $\{y_1, ..., y_{t-1}\}$. The prediction model is typically a classifier over the vocabulary with, say, 30,000 words.

2.2 The Attention Mechanism

The attention mechanism was first introduced to Seq2Seq (Bahdanau et al., 2014) to release the burden of summarizing the entire source into a fixed-length vector as context. Instead, the attention uses a dynamically changing context $\mathbf{c}_t$ in the decoding process. A natural option (or rather "soft attention") is to represent $\mathbf{c}_t$ as the weighted sum of the source hidden states, i.e.

$$\mathbf{c}_t = \sum_{\tau=1}^{T_S} \alpha_{t\tau}\mathbf{h}_\tau; \quad \alpha_{t\tau} = \frac{e^{\eta(\mathbf{s}_{t-1}, \mathbf{h}_\tau)}}{\sum_{\tau'} e^{\eta(s_{t-1}, \mathbf{h}_{\tau'})}} \quad (3)$$

where η is the function that shows the correspondence strength for attention, approximated usually with a multi-layer neural network (DNN). Note that in (Bahdanau et al., 2014) the source sentence is encoded with a Bi-directional RNN, making each hidden state $\mathbf{h}_\tau$ aware of the contextual information from both ends.

3 COPYNET

From a cognitive perspective, the copying mechanism is related to rote memorization, requiring less understanding but ensuring high literal fidelity. From a modeling perspective, the copying operations are more rigid and symbolic, making it more difficult than soft attention mechanism to integrate into a fully differentiable neural model. In this section, we present COPYNET, a differentiable Seq2Seq model with "copying mechanism", which can be trained in an end-to-end fashion with just gradient descent.

3.1 Model Overview

As illustrated in Figure 1, COPYNET is still an encoder-decoder (in a slightly generalized sense). The source sequence is transformed by **Encoder** into representation, which is then read by **Decoder** to generate the target sequence.

Encoder: Same as in (Bahdanau et al., 2014), a bi-directional RNN is used to transform the source sequence into a series of hidden states with equal length, with each hidden state $\mathbf{h}_t$ corresponding to word x_t. This new representation of the source, $\{\mathbf{h}_1, ..., \mathbf{h}_{T_S}\}$, is considered to be a short-term memory (referred to as $\mathbf{M}$ in the remainder of the paper), which will later be accessed in multiple ways in generating the target sequence (decoding).

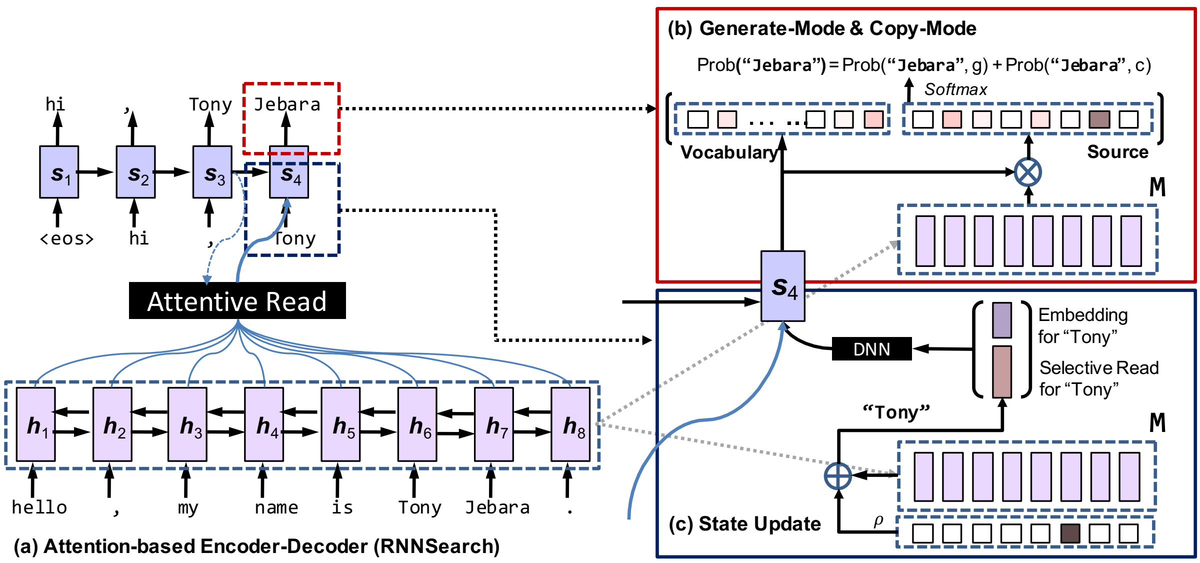

Figure 1: The overall diagram of COPYNET. For simplicity, we omit some links for prediction (see Sections 3.2 for more details).

Decoder: An RNN that reads $\mathbf{M}$ and predicts the target sequence. It is similar with the canonical RNN-decoder in (Bahdanau et al., 2014), with however the following important differences

- **Prediction:** COPYNET predicts words based on a mixed probabilistic model of two modes, namely the **generate-mode** and the **copy-mode**, where the latter picks words from the source sequence (see Section 3.2);
- **State Update:** the predicted word at time t–1 is used in updating the state at t, but COPYNET uses not only its word-embedding but also its corresponding location-specific hidden state in $\mathbf{M}$ (if any) (see Section 3.3 for more details);
- **Reading $\mathbf{M}$:** in addition to the attentive read to $\mathbf{M}$, COPYNET also has"selective read" to $\mathbf{M}$, which leads to a powerful hybrid of content-based addressing and location-based addressing (see both Sections 3.3 and 3.4 for more discussion).

3.2 Prediction with Copying and Generation

We assume a vocabulary $\mathcal{V} = \{v_1, ..., v_N\}$, and use UNK for any out-of-vocabulary (OOV) word. In addition, we have another set of words $\mathcal{X}$, for all the *unique* words in source sequence $X = \{x_1, ..., x_{T_S}\}$. Since $\mathcal{X}$ may contain words not in $\mathcal{V}$, copying sub-sequence in X enables COPYNET to output some OOV words. In a nutshell, the instance-specific vocabulary for source X is $\mathcal{V} \cup \text{UNK} \cup \mathcal{X}$.

Given the decoder RNN state $\mathbf{s}_t$ at time t together with $\mathbf{M}$, the probability of generating any target word y_t, is given by the "mixture" of probabilities as follows

$$p(y_t|\mathbf{s}_t, y_{t-1}, \mathbf{c}_t, \mathbf{M}) = p(y_t, \mathsf{g}|\mathbf{s}_t, y_{t-1}, \mathbf{c}_t, \mathbf{M}) + p(y_t, \mathsf{c}|\mathbf{s}_t, y_{t-1}, \mathbf{c}_t, \mathbf{M}) \quad (4)$$

where g stands for the generate-mode, and c the copy mode. The probability of the two modes are given respectively by

$$p(y_t, \mathsf{g}|\cdot) = \begin{cases} \frac{1}{Z} e^{\psi_g(y_t)}, & y_t \in \mathcal{V} \\ 0, & y_t \in \mathcal{X} \cap \bar{V} \\ \frac{1}{Z} e^{\psi_g(\text{UNK})} & y_t \notin \mathcal{V} \cup \mathcal{X} \end{cases} \quad (5)$$

$$p(y_t, \mathsf{c}|\cdot) = \begin{cases} \frac{1}{Z} \sum_{j: x_j = y_t} e^{\psi_c(x_j)}, & y_t \in \mathcal{X} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where $\psi_g(\cdot)$ and $\psi_c(\cdot)$ are score functions for generate-mode and copy-mode, respectively, and Z is the normalization term shared by the two modes, $Z = \sum_{v \in \mathcal{V} \cup \{\text{UNK}\}} e^{\psi_g(v)} + \sum_{x \in X} e^{\psi_c(x)}$. Due to the shared normalization term, the two modes are basically competing through a softmax function (see Figure 1 for an illustration with example), rendering Eq.(4) different from the canonical definition of the mixture model (McLachlan and Basford, 1988). This is also pictorially illustrated in Figure 2. The score of each mode is calculated: